

Concentration of Measure under Diffeomorphisms: What Coordinate Choice Can and Cannot Do

(corrected and rescoped version)

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Abstract

We revisit the idea that concentration of measure can be “improved” by an optimal choice of coordinate (diffeomorphism). The transport (covariance) of concentration inequalities under a diffeomorphism ϕ is exact and useful as an organizing language. However, the stronger claims — a universal optimal coordinate, and order-of-magnitude “strict improvement” over classical inequalities — do not survive scrutiny: the proposed concentration functional is not scale-normalized, its unconstrained minimization over the diffeomorphism group is degenerate (infimum 0), and the advertised improvement factors compare quantities carried in different units (and, in fact, bounds on *different* functionals). We give the corrected statements. The genuine residue is twofold: (i) for a *fixed* target functional the bounded-difference bound is coordinate-intrinsic, so the only legitimate role of ϕ is to *match the coordinate to the target* (e.g. the logarithm for multiplicative data), turning an inapplicable inequality into an applicable one — a *qualitative* (finite-vs-infinite) gain for heavy-tailed/multiplicative data, not a numerical beat on the same quantity; and (ii) the canonically scale-invariant concentration object is the Hellinger affinity, multiplicative under products, whose $-\log$ is additive — i.e. concentration of evidence lives in the logarithmic twin coordinate, where the Barron–Cover/Kraft bound is the natural large-deviation estimate. A closing section determines precisely what these corrected results do, and do not, contribute to orbital/MDL statistical estimation.

Contents

1	Introduction and the corrected thesis	1
2	The covariance dictionary (correct)	2
3	No free improvement: the central correction	2
4	Multiplicative data: the matched coordinate (corrected)	3
5	Information-geometric coordinate (corrected)	4
6	The scale-invariant object: affinity concentration and the link to estimation	4
7	Honest scope: what is, and is not, a contribution	5

1 Introduction and the corrected thesis

Let $(E, \mathcal{A})$ carry a probability measure μ , and let $\phi : E \rightarrow F$ be a diffeomorphism (or measurable bijection) with pushforward $\nu = \phi_*\mu$. The *transport principle* is the elementary identity

$$\int_E (g \circ \phi) d\mu = \int_F g d\nu, \quad \text{Law}_\mu(g \circ \phi) = \text{Law}_\nu(g). \quad (1)$$

Thus any concentration statement for the pair (g, ν) is *the same* statement for the pair $(g \circ \phi, \mu)$. This is correct and is a convenient language, but it is an isomorphism: it relabels, it does not create.

The temptation is to treat ϕ as a free knob and to “optimize” concentration over it. We show this is ill-posed as usually stated and that the publicized gains are artifacts (Section 3). We then isolate the two things that are genuinely true and useful:

- (i) **Match the coordinate to the target.** For a *fixed* functional the bounded-difference / sub-Gaussian constants are intrinsic; the role of ϕ is only to express the target where a classical inequality *applies*. For heavy-tailed or multiplicative data this is a *qualitative* gain (a finite bound where none existed), not a numerical improvement on the same quantity (Sections 4–5).
- (ii) **The scale-invariant object is the affinity.** The Hellinger affinity is multiplicative under products, so $-\log$ of it is additive: evidence concentrates in the logarithmic (twin) coordinate, where the Barron–Cover/Kraft bound is the natural large-deviation estimate (Section 6).

2 The covariance dictionary (correct)

Definition 2.1 (Pushforward, pullback). For $\phi : E \rightarrow F$ and μ on E , the pushforward is $(\phi_*\mu)(B) = \mu(\phi^{-1}(B))$; for $g : F \rightarrow \mathbb{R}$ the pullback is $\phi^*g = g \circ \phi$; for a metric d_F the pullback metric is $(\phi^*d_F)(x, y) = d_F(\phi(x), \phi(y))$.

Lemma 2.2 (Exact covariance of concentration). *Let $\nu = \phi_*\mu$. Then for every functional g , $\text{Law}_\mu(g \circ \phi) = \text{Law}_\nu(g)$. Consequently:*

- (a) $g \circ \phi$ is sub-Gaussian under μ with parameter σ^2 iff g is sub-Gaussian under ν with the same parameter σ^2 ;
- (b) ν satisfies a log-Sobolev (resp. Poincaré) inequality with constant C iff μ satisfies the pulled-back inequality with the same constant C ;
- (c) the isoperimetric profile is covariant: $I_\mu^{\phi^*d} = I_\nu^d$.

Proof. Immediate from (1): entropy, variance, Lipschitz seminorms and boundary measures are computed through integrals that are invariant under the change of variables $x \mapsto \phi(x)$. $\square$

Remark 2.3 (What Lemma 2.2 does *not* say). It transports a *single* inequality between a measure and its pushforward. It does *not* unify the distinct classical inequalities (Hoeffding, Bernstein, Talagrand, Gaussian isoperimetry): these rest on different hypotheses (bounded range, variance control, convexity, Gaussianity) and are not coordinate images of one another. The earlier claim that they are “manifestations of a single principle” over-reaches.

3 No free improvement: the central correction

Fix the sub-Gaussian functional of a measure,

$$F[\mu, \phi] := \inf\{\sigma^2 : \phi_*\mu \text{ is sub-Gaussian with parameter } \sigma^2\}. \quad (2)$$

Proposition 3.1 (The unconstrained optimum is degenerate). *For any non-degenerate μ on $\mathbb{R}^n$,*

$$\inf_{\phi \in \text{Diff}(\mathbb{R}^n)} F[\mu, \phi] = 0,$$

and the infimum is not attained by a diffeomorphism: it is approached by compressive maps $\phi = \varepsilon\psi$ (ψ bounded, $\varepsilon \rightarrow 0$), or by any ϕ with ϕ_μ supported in a vanishing ball. Hence “ $\phi^* = \arg \min_{\phi} \sup_{\mu} F[\mu, \phi]$ ” is ill-posed without a normalization of ϕ .*

Proof. $F[\varepsilon\psi \cdot; \mu] = \varepsilon^2 F[\psi_*\mu] \rightarrow 0$. A deviation of $\phi(X)$ by t corresponds to a deviation of X by $\asymp t/\varepsilon \rightarrow \infty$; the small “concentration constant” is bought by inflating the scale of the quantity being controlled. $\square$

The correct way to make the question meaningful is to *fix what is being bounded*.

Proposition 3.2 (Bounded-difference bounds are coordinate-intrinsic). *Let $h : E^n \rightarrow \mathbb{R}$ be a fixed functional with bounded differences c_i : for all x, x'_i , $|h(\dots, x_i, \dots) - h(\dots, x'_i, \dots)| \leq c_i$ whenever x_i, x'_i range over the support. Then $P(|h - \mathbb{E}h| \geq t) \leq 2 \exp(-2t^2 / \sum_i c_i^2)$, and the constants c_i depend only on h and the supports, not on any coordinate ϕ used to describe the data. In particular no diffeomorphism improves the McDiarmid bound for a fixed h .*

Proof. McDiarmid’s inequality applied to h . Reparametrizing the data by ϕ replaces h by $h \circ \phi^{-1}$ acting on $\phi(X_i)$; the increments are unchanged because $h \circ \phi^{-1}(\phi(x)) = h(x)$, so c_i is the same number computed in either chart. $\square$

Remark 3.3 (Consequence for “optimal coordinates”). Propositions 3.1–3.2 together say: the only legitimate effect of ϕ is to change *which functional* one bounds (or to make a previously inapplicable inequality applicable). There is no universal coordinate that tightens concentration for a fixed target; the well-posed problem is to *match* ϕ to the target functional.

4 Multiplicative data: the matched coordinate (corrected)

Let $X_1, \dots, X_n$ be independent with $X_i \in [a_i, b_i] \subset (0, \infty)$, and let $P_n = \prod_i X_i$. The natural (scale-invariant) deviation of a product is *multiplicative*, i.e. a deviation of $\log P_n$.

Theorem 4.1 (Concentration of products, corrected).

$$P(|\log P_n - \mathbb{E} \log P_n| \geq t) \leq 2 \exp\left(-\frac{2t^2}{\sum_{i=1}^n \log^2(b_i/a_i)}\right).$$

Equivalently, the multiplicative deviation $P_n / \exp(\mathbb{E} \log P_n)$ is controlled at the dimensionless scale $\sum_i \log^2(b_i/a_i)$.

Proof. $\log P_n = \sum_i \log X_i$ with $\log X_i \in [\log a_i, \log b_i]$ of range $\log(b_i/a_i)$; apply Hoeffding. By Proposition 3.2 the same constant is obtained by McDiarmid on the functional $\log P_n$ in *any* chart: the logarithm is merely the chart in which $\log P_n$ is a sum of bounded terms. $\square$

Remark 4.2 (Why there is no “ $\Gamma \approx 21000$ ” free lunch). The previously claimed improvement factor $\Gamma = (b - a)^2 / (\log(b/a))^2$ compares two *different* objects: the Hoeffding constant for the *sum* $S_n = \sum_i X_i$ (in units of X^2) against the Hoeffding constant for the *product* P_n (dimensionless). The numerator is scale-dependent (under $X \mapsto \lambda X$ it is multiplied by λ^2), the denominator is not, so Γ can be made arbitrary by rescaling and is not a function of b/a alone. For one *fixed* functional there is no such factor (Proposition 3.2).

Theorem 4.3 (Qualitative gain for heavy tails). *Suppose $X > 0$ is heavy-tailed: it admits no finite sub-Gaussian (or Bernstein-type) parameter in the identity chart, while $\log X$ is bounded/sub-Gaussian. Then no Hoeffding/Bernstein bound exists for X directly, whereas Theorem 4.1 and Lemma 2.2 yield finite concentration for the multiplicative functionals of X (products, geometric means, quantiles). The improvement is qualitative (finite versus infinite), not a numerical ratio on a common quantity.*

Proof. Direct from Lemma 2.2: sub-Gaussianity of $\log X$ transports to all functionals expressible through $\log X$; no statement about X in the identity chart is implied or claimed. $\square$

Theorem 4.4 (Worst case over a bounded class, corrected). *Over all μ supported in $[a, b]$, the sub-Gaussian functional is maximized, not minimized, at the two-point (Rademacher) law on $\{a, b\}$:*

$$\sup_{\text{supp}(\mu) \subseteq [a, b]} F[\mu, \text{id}] = \frac{(b - a)^2}{4}, \quad \inf_{\text{supp}(\mu) \subseteq [a, b]} F[\mu, \text{id}] = 0 \text{ (Dirac)}.$$

Hence the class-uniform Hoeffding constant in the identity chart is $(b - a)^2/4$ (a sup), and in the logarithmic chart $(\log(b/a))^2/4$; these are worst-case constants for different functionals/units and are not directly comparable.

Proof. For a distribution on $[a, b]$, $\text{Var} \leq (b - a)^2/4$ with equality at the endpoints two-point law (Popoviciu/Bhatia–Davis), and $\text{Var} \geq 0$ with equality at a Dirac. The earlier statement labelled this extremum an “inf”; it is a sup. $\square$

5 Information-geometric coordinate (corrected)

Proposition 5.1 (Expectation coordinate as the matched chart). *Let $P_\theta(dx) = \exp(\langle \theta, T(x) \rangle - A(\theta)) \mu_0(dx)$ be a regular exponential family with log-partition A . The expectation coordinate $\eta = \nabla A(\theta) = \mathbb{E}_\theta[T]$ is the canonical chart for the sufficient statistic: the Fisher information $\nabla^2 A(\theta) = \text{Var}_\theta(T)$ transforms into $\nabla^2 A^*(\eta)$ with A^* the Legendre dual, and the family is dually flat (the e - and m -connections are flat).*

Proof. Standard convex-duality computation; $\eta = \nabla A(\theta)$, $\theta = \nabla A^*(\eta)$, $\nabla^2 A^* = (\nabla^2 A)^{-1}$. $\square$

Remark 5.2 (Correction to “Fisher–Rao becomes Euclidean”). Dual flatness is *not* the same as making the Fisher–Rao (Levi-Civita) metric Euclidean. For $\dim \geq 2$ the Fisher–Rao metric of an exponential family generally has nonzero Riemannian curvature, so *no* chart renders it Euclidean; only the dual-affine structure flattens. The matched chart η is canonical in the dually-flat sense, which is the correct statement.

6 The scale-invariant object: affinity concentration and the link to estimation

The well-posed, scale-invariant concentration object is not $F[\mu, \phi]$ but the *Hellinger affinity*

$$\rho(P, Q) = \int \sqrt{dP dQ} \in (0, 1], \quad h^2(P, Q) = 1 - \rho(P, Q).$$

Proposition 6.1 (Evidence concentrates in the logarithmic twin coordinate). *The affinity is multiplicative under products, $\rho(P^{\otimes n}, Q^{\otimes n}) = \rho(P, Q)^n$, so $-\log \rho$ is additive. For a countable model family $\{f_m\}$ with prefix code lengths satisfying the half-Kraft condition $\sum_m 2^{-L(m)/2} \leq 1$, the minimum-description-length selector $\hat{f}$ obeys the partition-function (large-deviation) bound*

$$\mathbb{E}_{P^*} \left[2^{-L(\hat{m})/2} \prod_{i=1}^n \sqrt{\hat{f}(X_i)/f^*(X_i)} \right] \leq \sum_m 2^{-L(m)/2} \rho(f^*, f_m)^n \leq 1, \quad (3)$$

whence $\mathbb{E}[h^2(P^*, \hat{f})] \leq \inf_m \{\text{KL}(P^* \| f_m) + (\log 2)L(m)/n\}$.

Proof. Multiplicativity of ρ is Fubini; (3) is the Barron–Cover argument: $\hat{m}$ maximizes $2^{-L(m)} \prod_i f_m(X_i)$, take square roots and bound the max by the sum; then take $\mathbb{E}_{P^*}$. The resolvability bound follows by the standard passage (Jensen, $h^2 \leq -\log \rho \leq \text{KL}$). $\square$

This is the correct form of “concentration in a twin space”: the relevant non-additive structure is the *multiplicative* space of affinities/likelihood ratios, whose additive (logarithmic) chart is canonical and *scale invariant* — precisely the normalization that $F[\mu, \phi]$ lacks.

Remark 6.2 (Scope: near-minimax, not exact). The resolvability bound of Proposition 6.1 is the genuine contribution to orbital/MDL estimation, but it must be read precisely. On a nonparametric sieve the code length is $L(m) \asymp D_m \log n$, so the bound yields the rate $(\log n/n)^{2s/(2s+d)}$ — minimax up to a logarithmic factor, the per-coefficient quantization cost of the description-length code. The affinity/MDL machinery is thus *near-minimax*; the *exact* rate $n^{-2s/(2s+d)}$ is attained not by this code but by a projection / penalized-projection estimator paying the metric complexity D_m/n . What these corrected results contribute is the concentration step (the affinity bound), not the removal of the log, which is a separate, estimator-level refinement.

7 Honest scope: what is, and is not, a contribution

Retained / correct.

- The covariance dictionary (Lemma 2.2): a clean transport of sub-Gaussianity, log-Sobolev, Poincaré, isoperimetry between a measure and its pushforward.
- The matched-coordinate principle for multiplicative/heavy-tailed data (Theorems 4.1–4.3): a *qualitative* extension of applicability.
- The expectation/dually-flat chart for exponential families (Proposition 5.1).
- The affinity formulation (Proposition 6.1) as the scale-invariant concentration object.

Withdrawn / corrected.

- No universal optimal coordinate: $\inf_\phi F[\mu, \phi] = 0$ (Proposition 3.1).

- No order-of-magnitude “strict improvement”: the factor $(b-a)^2/(\log(b/a))^2$ mixes units and functionals (Proposition 3.2, Theorem 4.1).
- The internal inconsistency between a threshold $b/a = e^2$ and a uniformly ≥ 1 improvement factor is removed (there is no such factor for a fixed functional).
- Minimax inf corrected to sup (Theorem 4.4).
- No unification of Hoeffding/Bernstein/Talagrand/Gaussian isoperimetry; no chart making Fisher–Rao Euclidean.

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